rewrite Eq. (5) as

$$\vec{r}_1^{(z_{1:\mathrm{d}})} = \boldsymbol{\pi} + \sum_{\mathrm{s}=1}^{\mathrm{d}} \sum_{\lambda \in \Lambda_T \setminus \{1\}} \lambda^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_\mathrm{s}} T_\lambda \tag{7}$$

where T_λ is the spectral projection operator associated with eigenvalue λ (Riechers & Crutchfield, 2018a). In this diagonalizable case, $T_\lambda = \sum_{k=1}^{a_\lambda} |\lambda_k\rangle\langle\lambda_k|$, where a_λ is the algebraic multiplicity of the eigenvalue λ, with right eigenstates satisfying $T|\lambda_k\rangle = \lambda|\lambda_k\rangle$, left eigenstates satisfying $\langle\lambda_k| T = \lambda\langle\lambda_k|$, all satisfying the orthonormality condition $\langle\lambda_j|\lambda_k\rangle = \delta_{j,k}$. Notably in Eq. (7), all dependence on inter-token distance now lies solely in the exponentiation of the eigenvalues, which all live on or within the unit circle in the complex plane for a stochastic transition matrix like T.

For the Mess3 process, the stochastic matrix T has eigenvalues $\Lambda_T = \{1, \zeta\}$, where $\zeta = 1 - 3x$ is a degenerate eigenvalue with multiplicity $a_\zeta = 2$. We observed that the attention weight n tokens back is approximately $\zeta^n = (1-3x)^n$, which suggests a strong connection between the theoretically motivated Eq. (7) and the architectural-implementation Eq. (3). Encouraged by this correspondence and further evidence of similarity, we make the ansatz that *the role of attention in the first layer is to implement the constrained belief update of Eq.* (6) *via Eq.* (7)*'s spectral mechanism* [2]. Taking this ansatz seriously allows us to precisely anticipate the analytic form of the learned attention pattern.

To derive the analytic form of the attention pattern, we assume that there is a linear map $f : \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{|\mathcal{S}|-1}$ from the residual stream to the hyperplane containing the probability simplex over the hidden states of a minimal generative model of the data (the 2-simplex in this case). Let $\Pi_{\boldsymbol{\Delta}} = I - T_1 = I - \mathbf{1}\boldsymbol{\pi}$ be the projection from $\mathbb{R}^{|\mathcal{S}|}$ to the hyperplane $\mathbb{R}^{|\mathcal{S}|-1}$ containing the simplex. Our full ansatz is thus $f(\vec{x}_\mathrm{d}^{(\mathrm{mid})}) = \vec{r}_1^{(z_{1:\mathrm{d}})}\Pi_{\boldsymbol{\Delta}}$ or, more explicitly:

$$f(\vec{x}_\mathrm{d}^{(\mathrm{mid})}) = \sum_{\mathrm{s}=1}^{\mathrm{d}} \sum_{\lambda \in \Lambda_T \setminus \{1\}} \lambda^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_\mathrm{s}} T_\lambda \tag{8}$$

$$= f(\vec{x}_\mathrm{d}^{(\mathrm{pre})}) + \sum_{\mathrm{s} \leq \mathrm{d}} A_{\mathrm{d},\mathrm{s}} f(\vec{v}_\mathrm{s}) \,. \tag{9}$$

From this, we group source-specific terms to infer that

$$f(\vec{x}_\mathrm{d}^{(\mathrm{pre})}) + A_{\mathrm{d},\mathrm{d}} f(\vec{v}_\mathrm{d}) = \boldsymbol{\pi} T^{|z_\mathrm{d}} - \boldsymbol{\pi} \tag{10}$$

and

$$A_{\mathrm{d},\mathrm{s}} f(\vec{v}_\mathrm{s}) = \sum_{\lambda \in \Lambda_T \setminus \{1\}} \lambda^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_\mathrm{s}} T_\lambda \quad \text{for } \mathrm{d} > \mathrm{s} \,. \tag{11}$$

From Eq. (11), we notice that $f(\vec{v}_\mathrm{s})$ is in the linear span of the non-stationary left eigenstates of T. I.e., $f(\vec{v}_\mathrm{s}) \in \mathrm{span}\big(\{\langle\lambda| : \lambda\langle\lambda| = T\langle\lambda| \text{ and } \lambda \neq 1\}\big)$ and, in particular, $f(\vec{v}_\mathrm{s}) \cdot |1\rangle = 0$ such that *adding any of the OV vectors to any stochastic vector (whose elements by definition add to one) keeps you in the hyperplane of the probability simplex*.

For the Mess3 family of processes, T has a single eigenvalue $\zeta = 1 - 3x$ with multiplicity $a_\zeta = 2$ besides its eigenvalue of 1. Accordingly, Eq. (11) simplifies to

$$A_{\mathrm{d},\mathrm{s}} f(\vec{v}_\mathrm{s}) = \zeta^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_\mathrm{s}} T_\zeta \qquad \text{for } \mathrm{d} > \mathrm{s} \,, \tag{12}$$

which forces $f(\vec{v}_\mathrm{s}) = c\boldsymbol{\pi} T^{|z_\mathrm{s}} T_\zeta$ for some $c \in \mathbb{R}$ independent of d, from which we obtain

$$A_{\mathrm{d}+m,\mathrm{s}} = \zeta^m A_{\mathrm{d},\mathrm{s}} \qquad \text{for } \mathrm{d} > \mathrm{s} \,. \tag{13}$$

So, for example, $A_{2,1}$ implies $A_{\mathrm{d},1}$ for all destinations d ≥ 2; and $A_{3,2}$ implies $A_{\mathrm{d},2}$ for all destinations d ≥ 3.

For Mess3, $T_\zeta = I - |1\rangle\langle 1| = I - \mathbf{1}\boldsymbol{\pi}$, since all projection operators must sum to the identity. Combining this insight with Eq. (12) tells us about the OV-vector for all positions:

$$f(\vec{v}_m) = \frac{\zeta}{A_{m+1,m}} \big(\boldsymbol{\pi} T^{|z_m} - \boldsymbol{\pi}\big) \,. \tag{14}$$

Notably, Eq. (14) tells us that all OV-vectors associated with the same token must be parallel—$f(\vec{v}_\mathrm{s}) \propto f(\vec{v}_{\mathrm{s}'})$ if $z_\mathrm{s} = z_{\mathrm{s}'}$—which is consistent with what we observe in our experiments (Fig. 2A). Moreover, the magnitude of the mth OV-vector is inversely proportional to the attention element $A_{m+1,m}$, which is again consistent with our experiments (Fig. 2AB). In our experiments, we find $A_{2,1}$ to be significantly larger than all the other $A_{m+1,m}$ elements, while the latter all cluster together; the magnitude of $\vec{v}_1$ is correspondingly smaller than all of the other strongly clustered $\vec{v}_m$ magnitudes.

Combining Eqs. (10) and (14) constrains the embedding

$$f(\vec{x}_m^{(\mathrm{pre})}) = \Big(1 - \tfrac{\zeta A_{m,m}}{A_{m+1,m}}\Big)\big(\boldsymbol{\pi} T^{|z_m} - \boldsymbol{\pi}\big) \tag{15}$$

to be parallel to the OV-vectors, as we indeed observe.

Eqs. (13), (14) and (15) make *strong predictions about the form of the attention pattern and how it relates to OV-vectors and token embeddings*, which must be true if the first layer of attention is indeed implementing the constrained belief updates over latent states of a generative model of the training data. These relationships are all borne out in our experiments (Fig. 2ABC), except for some scalar discrepancy in the first two embedding vectors (see Appendix D for quantification), which is a strong validation of the predictive power of our framework.

4.4.1. Negative eigenvalues require more attention heads

For the transition matrix T to be row stochastic (a requirement for a valid HMM), x must be in the range $[0, 0.5]$.

[2] The details of this correspondence break down if there are many attention heads in the first layer.

Interestingly, when $\zeta < 0$ (which occurs when $x > 1/3$), the predicted pattern oscillates and cannot be captured by a single attention head, since attention pattern entries must be non-negative. In these cases, we observe that a single-head transformer captures an incomplete representation of the belief state geometry, and the transformer performs correspondingly worse (App. F). However, upon adding a second attention head, the model converges to the solution predicted by the belief updating equation, even in the presence of oscillatory dynamics, as shown in Fig. 3.

The anticipated need for a second attention head when the data-generating transition matrix has a negative eigenvalue further demonstrates how our analysis provides a handle to relate the architectural constraints of the attention mechanism to the structure of the training data. In fact, our framework provides more specific predictions for the attention pattern and its relation to embedding and OV-vectors in this case too.

With two attention heads, the update to the residual stream at the destination position s becomes

$$\vec{c}_{\mathrm{d}} = \sum_{\mathrm{s}=1}^{\mathrm{d}} \sum_{h=1}^{2} A_{\mathrm{d,s}}^{(h)} \vec{v}_{\mathrm{s}}^{(h)} \,, \tag{16}$$

where each head now has its own QK and OV matrices. With the negative eigenvalue $\zeta < 0$ and two attention heads, we can relate the constrained belief update to the details of attention and embedding via

$$\begin{aligned} f(\vec{x}_{\mathrm{d}}^{(\mathrm{mid})}) &= \sum_{\mathrm{s}=1}^{\mathrm{d}} (-1)^{\mathrm{d}-\mathrm{s}} (-\zeta)^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_{\mathrm{s}}} T_{\zeta} \qquad (17) \\ &= f(\vec{x}_{\mathrm{d}}^{(\mathrm{pre})}) + \sum_{\mathrm{s} \leq \mathrm{d}} \left[A_{\mathrm{d,s}}^{(1)} f(\vec{v}_{\mathrm{s}}^{(1)}) + A_{\mathrm{d,s}}^{(2)} f(\vec{v}_{\mathrm{s}}^{(2)}) \right] . \end{aligned}$$

This is naturally accommodated by

$$A_{\mathrm{d,s}}^{(1)} f(\vec{v}_{\mathrm{s}}^{(1)}) = +\delta_{+1,(-1)^{\mathrm{d}-\mathrm{s}}} |\zeta|^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_{\mathrm{s}}} T_{\zeta} \quad \text{and} \tag{18}$$

$$A_{\mathrm{d,s}}^{(2)} f(\vec{v}_{\mathrm{s}}^{(2)}) = -\delta_{-1,(-1)^{\mathrm{d}-\mathrm{s}}} |\zeta|^{\mathrm{d}-\mathrm{s}} \boldsymbol{\pi} T^{|z_{\mathrm{s}}} T_{\zeta} \tag{19}$$

for $\mathrm{d} > \mathrm{s}$, which implies that the OV-vectors point in opposite directions, $\widehat{f(\vec{v}_{\mathrm{s}}^{(1)})} = -\widehat{f(\vec{v}_{\mathrm{s}}^{(2)})}$, with $f(\vec{v}_{\mathrm{s}}^{(h)}) \propto (\boldsymbol{\pi} T^{|z_{\mathrm{s}}} - \boldsymbol{\pi})$ and

$$A_{\mathrm{d}+2m,\mathrm{s}}^{(h)} = \zeta^{2m} A_{\mathrm{d,s}}^{(h)} \qquad \text{for } \mathrm{d} > \mathrm{s} \,, \tag{20}$$

consistent with our experiments as shown in Fig. 3. We note that the magnitudes of OV vectors are tied to attention magnitudes via $c\zeta^{\mathrm{d}-\mathrm{s}} = A_{\mathrm{d,s}}^{(1)} |f(\vec{v}_{\mathrm{s}}^{(1)})| - A_{\mathrm{d,s}}^{(2)} |f(\vec{v}_{\mathrm{s}}^{(2)})|$, with $c = |\boldsymbol{\pi} T^{|z_{\mathrm{s}}} - \boldsymbol{\pi}| \in \mathbb{R}$, which is also observed in Fig. 3.

4.5. Post-MLP geometries

While the intermediate geometry is well characterized by our constrained belief equations, the MLP transformation is more complex. Through purely local computations at each position, the MLP learns a continuous nonlinear warping that transforms the intermediate fractal structure into the final belief geometry.

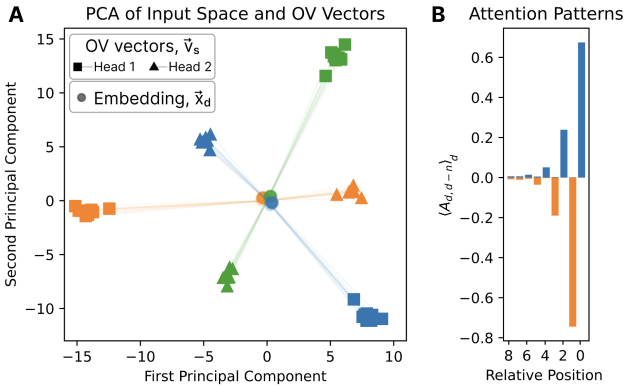

Figure 3: Attention heads combine to capture oscillatory dynamics in belief updating. (A) In the token embedding space, the model uses each attention head to embed tokens on opposite poles of the simplex. (B) The attention patterns of the two heads (shown here averaged over all sequences) act as positive and negative components. When combined, they produce the oscillatory pattern predicted by the exponentiated eigenvalue $\zeta^n = (1-3x)^n = (-1)^n(3x-1)^n$.

Fig. 4 shows the close match between theoretical predictions and observed representations across different Mess3 parameter settings, confirming our theoretical understanding. The transformation involves stretching and compressing different regions of the space while maintaining topological structure, though a full characterization of its mathematical properties remains for future work.

To verify that our single-layer analysis captures the fundamental computation even in deeper networks, we analyzed 4-layer transformers trained on the same Mess3 data. We find that the first attention layer consistently implements the constrained belief update mechanism, Eq. (5), with subsequent layers progressively transforming the representation toward the full Bayesian belief geometry, Eq. (2). Quantitative MSE analysis (Fig. A6) shows that first-layer attention outputs align well with constrained beliefs while later layers systematically improve alignment with full Bayesian beliefs. This progression is shown in Fig. A5, confirming that our theoretical framework for the first layer remains the foundational computation in multi-layer networks.

5. Discussion and Conclusion

We have shown how combining **computational mechanics** with **mechanistic interpretability** yields a principled understanding of why transformers trained on Mess3 HMM

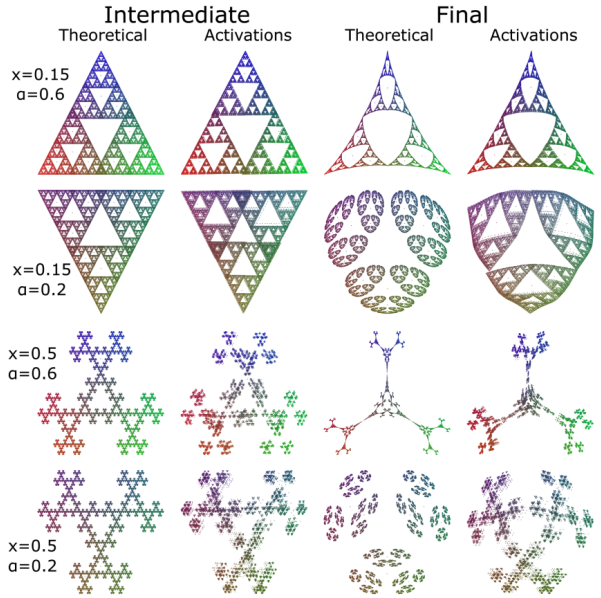

Figure 4: Comparison of model representations and theoretical predictions for different Mess3 hyperparameters in each row. Each subfigure shows four columns: (i) Intermediate representation from Eq. (5). (ii) PCA projection of the model activations in the intermediate layer. (iii) Ground truth belief state geometry from Eq. (2). (iv) PCA projection of the final activations after the MLP.

data learn intermediate fractal-like structures, and how these structures systematically transition into final belief-state representations. By focusing primarily on single-layer transformers, we isolated the fundamental computation performed by the attention mechanism, while also demonstrating that these principles extend to the first layer of deeper networks. This approach provides a top-down theoretical explanation grounded in the tension between optimal Bayesian belief updates and the parallel, attention-based constraints of transformer computation, developing geometric observations of activation space into mechanistic understanding of the underlying computational principles.

Implications for interpretability. Our work demonstrates an alternative to bottom-up architectural analysis. Knowing the structure of optimal predictors allows us to predict and verify the *specific* intermediate computations that are implemented by the attention mechanism. Our analysis reveals the computational role of specific directions in activation space—showing how the geometry of belief updates shapes the learned representations. Additionally, by focusing on a small, tractable HMM, we see how specific properties of its transition matrix lead to oscillatory patterns that require specialized multi-head solutions due to the non-negativity constraints of attention mechanisms. Rather than relying on general observations that attention heads specialize, our analysis reveals precisely *why* and *how* multiple heads must coordinate: the non-negativity constraints of attention, combined with oscillatory patterns in optimal belief updates, necessitate specific decompositions across heads, providing concrete mechanistic understanding of their functional roles. This demonstrates how combining theoretical understanding with architectural constraints can yield precise, verifiable interpretations of neural network components.

Limitations and future work. Our experiments used small transformers (1 and 4 layers), with analysis focused on the first attention layer. For training data we used the specialized Mess3 family of HMMs with full support over the space of all possible sequences of tokens. We discovered how transformers implement belief updates when attention patterns depend primarily on positional distances, while token-specific information is handled through value vectors. While we validated that the first layer of multi-layer transformers implements the same constrained belief updating mechanism (Figs. A5;A6), with subsequent layers refining toward full Bayesian beliefs, our techniques must be adapted to both larger transformer architectures and data-generating processes that capture the complexities of real-world data. While this setting offers clear insights, it does not capture many aspects of natural language. Future work could apply these techniques to processes that better reflect properties of natural language—hierarchical, with sparse support over sequences. Moreover, the interplay between multi-head attention and deeper layer stacks likely exhibits additional nuances that our primary single-layer analyses only begins to uncover. Finally, while we showed that the final MLP layer refines partial updates to approximate full Bayes, the deeper question of why gradient descent converges on these circuits remains ripe for further investigation.

Conclusion. By combining computational mechanics with mechanistic interpretability, we have shown how transformers implement inherently recursive Bayesian updates through parallel computations via the attention mechanism, and how these intermediate representations are refined into the final form. This reconciles model-agnostic theories of next-token prediction with the reality of architecture-specific constraints. We hope our results not only advance interpretability for HMM-like toy tasks but also inspire deeper theoretical insights into how large-scale transformers produce—and exploit—belief-like structures in real-world applications.

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